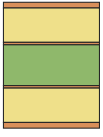
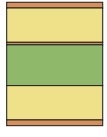
JLCPCB 4-layer



Σ 0.771

7628; εr 4.4/4.23/4.4

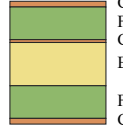
This work



Σ 0.756

L3 empty; εr 4.4/4.23/4.4

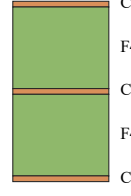
Wen 2021



Σ 0.753

εr 4.8 (one FR-4 value)

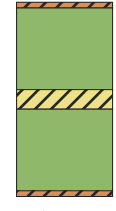
Wang 2022



Σ 1.105

εr 2.55

Fang 2025



Σ n/r

εr 2.55; inner Cu n/r

Lu 2026

Σ 0.905

εr 4.4